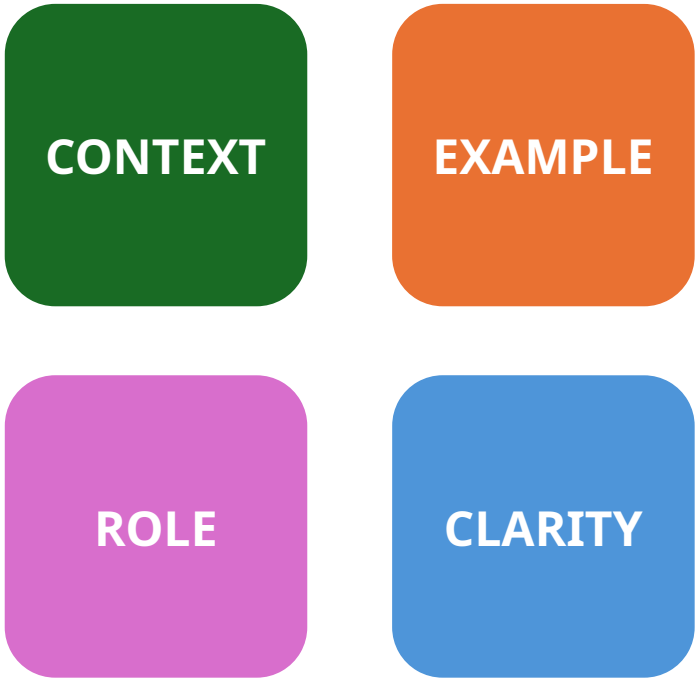
Advanced Prompt Engineering - II

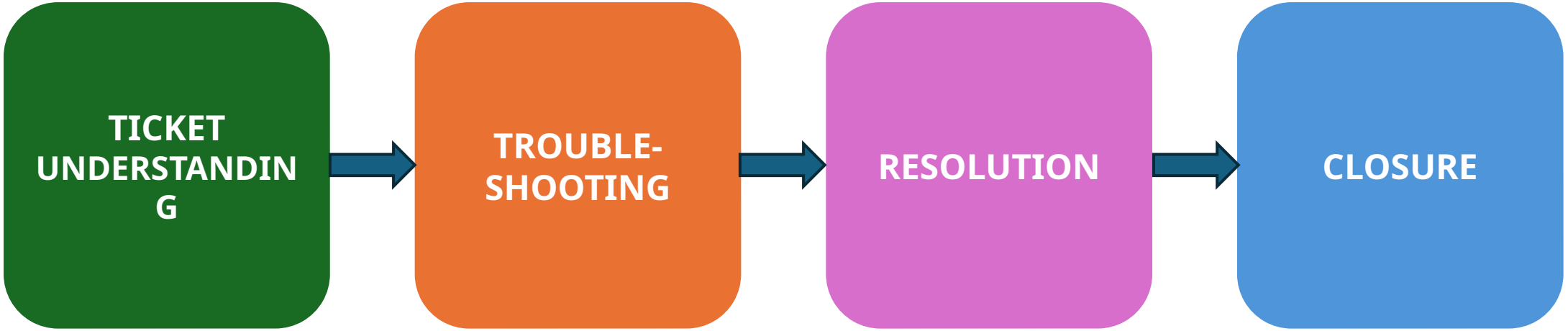
Session 2 – January 22nd, 2025

Quick Recap



1. **<u>Context:</u>** Always provide a background story for the problem at hand.
2. **<u>Examples:</u>** Provide examples - help model mimic
3. **<u>Assign Role:</u>** Provide a role to the model to guide and tailor its response.
4. **<u>Clarity:</u>** Clearly define what you want the Copilot to do.

Quick Recap


TICKET UNDERSTANDIN G
TROUBLE-SHOOTING
RESOLUTION
CLOSURE

Quick Recap

Text Extraction:
Identify key details from customer tickets.

Text Summarization:
Summarize long, complex tickets for quick understanding.

Text Generation:
Generate clear, empathetic responses or troubleshooting steps.

Knowledge Retrieval:
Retrieve relevant solutions from internal knowledge bases.

Anomaly Detection:
Identify unusual patterns or issues in tickets.

Multilingual Support:
Break language barriers with real-time translations.

Sentiment Analysis:
Understand customer tone and urgency.

Quick Recap

Stage	Capability	Outcomes
Ticket Understanding	Text Extraction, Sentiment Analysis	Extract version details; Identify negative sentiment.
Troubleshooting	Knowledge Retrieval, Anomaly Detection	Identify root causes from logs or patterns.
Resolution	Text Generation, Multilingual Support	Provide step-by-step guidance; Translate responses.
Closure	Sentiment Analysis, Text Summarization	Ensure satisfaction with follow-up communication.

Few Best Picks from Take Home

I am a senior virtual machine support engineer with 5 years of experience in windows at Microsoft.

I am troubleshooting an urgent issue for a customer that is unable to log in remotely. They are using a Windows Server 2022 hosted in the Azure West Europe region size Standard_B2ms to Standard_D2s_v3. The error they receive when logging in is Remote Desktop can’t connect to the remote computer for one of these reasons:

Remote access to the server is not enabled.
The remote computer is turned off.
The remote computer is not available on the network.”*

We have already tried the steps of making sure the network security group allows remote desktop on port 3389.

What additional firewall settings will block access to RDP? Provide additional troubleshooting steps for this error message.

Few important points :

- Please keep your Copilot open and accessible
- Be ready to take part in polls & quizzes
- Be ready to work on hands-on exercises

Training Outline

Module 1
(Week 1 : 2 Hours)

Crafting effective & Contextual Prompts

→

Module 2
(Week 2 : 2.5 Hours)

Refinement with Chaining & Advance Chaining Techniques

→

Module 3
(Week 3 : 3 Hours)

Probing, Empathy, Proactivity & Ethics

- Total Duration : 8 Hours
- 3 Modules in 3 Weeks
- Focused hand’s-on sessions

2

- Prompt Library
- Refinement with CoT
- Break
- Hands-on Practice
- Q&A
- Sequential, Conditional & Iterative Prompting
- Break
- Q&A
- Building Prompt Templates
- Q&A

"Have you used a structured, step-by-step approach in writing Copilot prompts?”

A. Yes, frequently.

B. Occasionally, but not consistently.

C. I’ve heard of it but haven’t used it.

D. This is the first time I’m learning about it.

P1

60 Seconds

“What is the most common issue you’ve faced when using Copilot for troubleshooting?"

A. Vague or incomplete responses.

B. Oversimplified answers.

C. Lack of logical reasoning.

D. Misleading or fabricated information.

P2

60 Seconds

Simple Prompt

Contain a single instruction or question for the model to respond to.

Complex Prompt

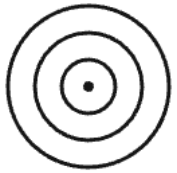
These prompts contain multiple instructions or questions that require the model to perform a series of actions or provide a detailed response.

Example : *"I'm looking for a restaurant that serves vegan food and is open until 10 pm. Can you recommend one?"*

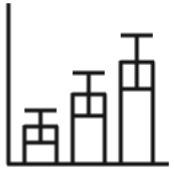
“A prompting technique that encourages the model to reason step-by-step, mimicking human logical thought processes.”

Chain of Thought

1. Consistency



2. Enhanced control



3. Reduced Error Rate

Advantages of CoT

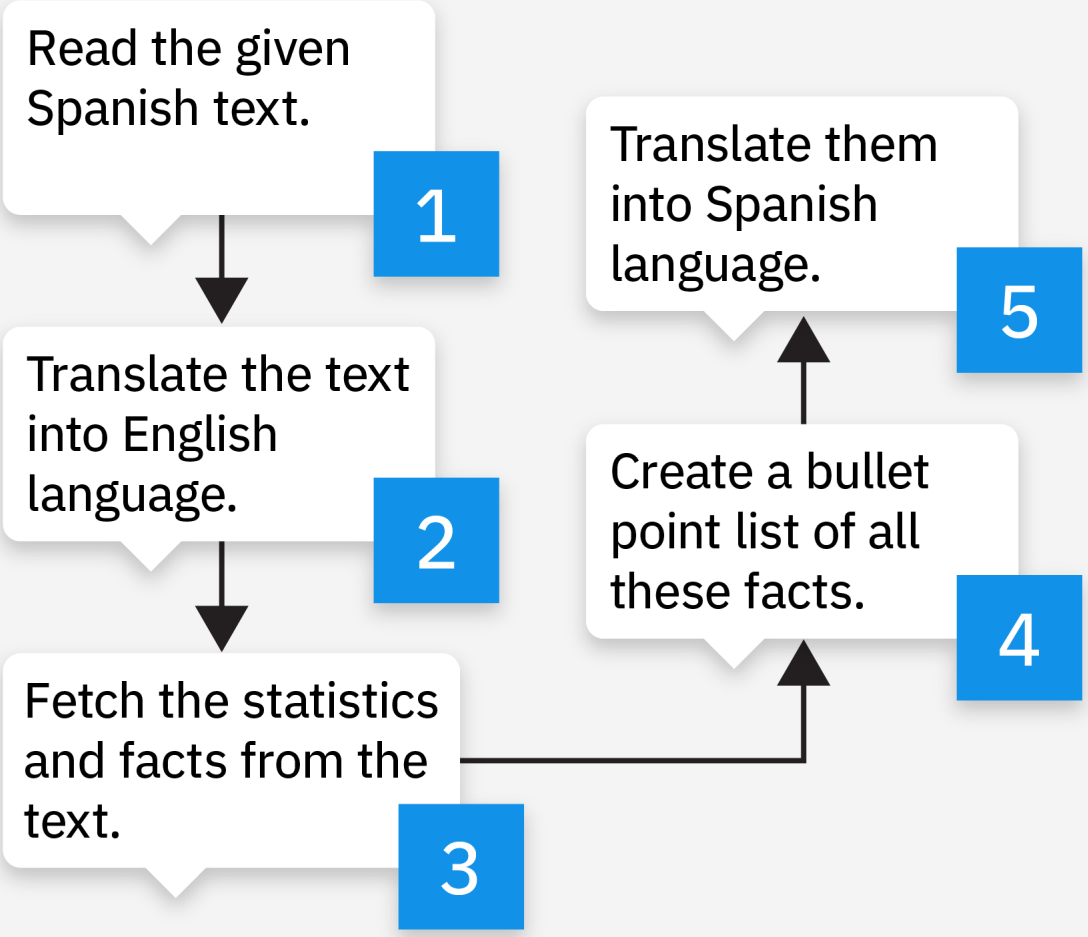
Complex Prompt

“Consider the given text in Spanish. Translate it into English. Find all the statistics and facts used in this text and list them as bullet points. Translate them again into Spanish.”

Complex Prompt

Consider the given text in Spanish. Translate it into English. Find all the statistics and facts used in this text and list them as bullet points. Translate them again into Spanish.

Simple Prompt



4 STEP FRAMEWORK

1. Identify the main goal or objective of the prompt.
2. Break down the main goal into sub-tasks, that is, more specific actions or tasks.
3. Create a prompt for each specific action or task.
4. Ensure that each prompt is clear, concise, and unambiguous.

WHEN TO USE “CHAIN OF THOUGHT”?

- **For Multi-Step Troubleshooting:**

Use CoT when diagnosing complex issues, such as Azure VM connectivity failures or Power BI refresh errors, where multiple factors need to be examined systematically.

- **When the Problem Has Ambiguity:**

Apply CoT when the issue isn’t clearly defined, like intermittent system failures, to ensure all potential causes are considered.

WHEN TO USE “CHAIN OF THOUGHT”?

- **To Debug Multi-Dependency Systems:**

Use CoT for systems with interdependent components, such as ARM templates or virtual network configurations, to analyze each dependency.

- **For Root Cause Analysis:**

When identifying the underlying cause of recurring issues, CoT ensures a detailed breakdown to isolate the source effectively.

BALANCING CoT USAGE

1. **Is the task complex or multi-step?**
   Use CoT if yes.
2. **Does the problem require reasoning or logical steps or has dependent systems?**
   Use CoT if yes.
3. **Would step-by-step reasoning improve transparency Of output?**
   Use CoT if yes.
4. **Is speed or simplicity more critical than detailed reasoning?**
   Avoid CoT if yes.

"What is the primary advantage of the Chain of Thought technique?"

A. It provides generic responses to save time.

B. It ensures a faster response, even if incomplete.

C. It breaks down complex problems into manageable steps.

D. It avoids repetition in responses by skipping intermediate steps.

60 Seconds

"In which of the following situations is the Chain of Thought technique most beneficial?"

A. When the problem is straightforward and well-defined.

B. Troubleshooting a Power BI refresh error with multiple potential causes.

C. When asking for a definition of a technical term.

D. When summarizing a research paper.

60 Seconds

"Which task do you think benefits the most from the Chain of Thought technique?"

A. Diagnosing multi-layered technical issues (e.g., Azure connectivity)

B. Optimizing performance of systems (e.g., Power BI reports)

C. Identifying root causes of failures (e.g., deployment errors)

D. All of the above.

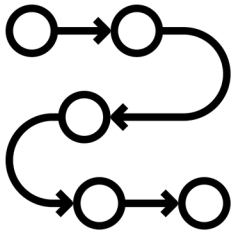
60 Seconds

Q & A

BREAK (7 MINS)

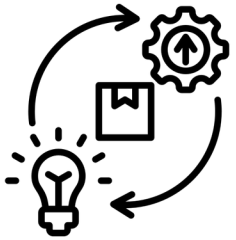
Hands On Session (15 Mins)

Advanced Chaining Techniques



Sequential Prompting

Conditional Prompting



Iterative Prompting

SEQUENTIAL PROMPTING

Sequential chaining involves linking prompts in a straightforward, linear sequence. Each prompt depends on the output of the previous one, creating a step-by-step flow of information and tasks.

Sequential Prompting

Test Customer Query

"Debugging an Azure Deployment Failure"

An Azure Resource Manager (ARM) template deployment has failed. The technical support engineer uses Copilot in a sequential process where the output of each step feeds into the next prompt.

Sequential Prompting

Step 1: Gather Basic Error Details

Prompt: "The ARM template deployment failed with an error: 'ResourceNotFound'. What initial details should I gather to begin troubleshooting?"

Copilot Response:

1. Identify the resource that caused the error (e.g., a storage account or VM).
2. Note the deployment region and subscription.
3. Collect the deployment log for detailed error messages.

Sequential Prompting

Step 2: Verify the missing resource

Prompt: "I found that 'storageAccount1' is missing in the 'East US' region. How can I verify if this resource exists in my Azure environment?”

Copilot Response:

1. Use the Azure CLI: az storage account list --query "[?name=='storageAccount1']" to check if it exists.
2. Confirm that the resource group containing this storage account is deployed in the correct region.
3. Review permissions to ensure the deployment has access to create or update this resource.

Step 3: Determine the Cause of the Missing Resource

Prompt: "The resource 'storageAccount1' does not exist in 'East US'. What are the common causes for this during an ARM template deployment?”

Copilot Response:

1. The template might reference an incorrect or nonexistent resource.
2. Dependencies for 'storageAccount1' might not be correctly specified in the template.
3. The deployment might target the wrong resource group or region.

Step 4: Fix the Template and Redeploy

Prompt: "I identified a typo in the ARM template for 'storageAccount1'. How do I correct it and ensure the deployment succeeds?”

Copilot Response:

1. Update the ARM template to reference the correct resource name and region.
2. Validate the template using az deployment group validate.
3. Redeploy the template with the corrected parameters.

Tasks with Defined Steps: Use when the problem requires a structured approach, such as system diagnostics or step-by-step troubleshooting.

Example: Analyzing Power BI performance by systematically checking data sources, DAX queries, visuals, and refresh schedules.

Comprehensive Coverage: Use when all potential factors of a problem must be investigated to ensure thoroughness.

Example: Debugging Azure VM network connectivity by examining NSG rules, VNet configurations, and IP settings.

Repeatable Processes: Use for workflows or procedures that need to follow a consistent sequence for quality assurance.

Example: Conducting a security audit or setting up a deployment pipeline.

When to Use Sequential Prompting

Dynamic or Branching Scenarios: Avoid when the next step depends on the output of the previous step.

Example: Diagnosing a server issue where one finding (e.g., disk failure) determines the next investigation area.

Simple or Straightforward Tasks: Avoid for tasks that don’t require multiple steps, as sequential prompting may unnecessarily complicate the process.

Example: Resetting a password or checking a single configuration.

Creative or Open-Ended Tasks: Avoid when brainstorming, storytelling, or solving problems with no fixed resolution path.

Example: Generating content ideas or designing user experiences.

When Not to Use Sequential Prompting

Sequential prompting is best suited for structured, repeatable, and comprehensive workflows but is less effective in dynamic, simple, or open-ended scenarios. It ensures thoroughness but lacks flexibility.

"Which of the following best represents Sequential Chaining in technical support?"

A. Investigating all potential causes of a problem simultaneously to save time.

B. Following a fixed series of steps to diagnose an issue, such as checking data source performance, optimizing DAX queries, and reviewing visuals for a slow Power BI report.

C. Adjusting troubleshooting steps dynamically based on each new finding.

D. Using AI to suggest random solutions for technical problems.

60 Seconds

CONDITIONAL PROMPTING

Conditional chaining introduces branching in the prompt chain, allowing flexible workflows where the next step adapts based on the LLM's output.

Test Customer Query

"Diagnosing a Power BI dataset refresh failure caused by a timeout error."

In conditional chaining, the next prompt depends dynamically on the output of the previous step. The sequence adapts based on the findings or outcomes at each stage, allowing the process to branch or adjust to new information.

Conditional Prompting

Step 1:

Define the Objective : "A Power BI dataset failed to refresh with a timeout error. What could be the possible causes?"

Copilot Response:

1. The data source might be unavailable or slow.
2. The gateway might be overloaded or misconfigured.
3. The dataset might contain long-running queries.

Conditional Prompting

Step 2:

Refine the Query : "Let’s investigate further. How can I confirm if the data source is causing the issue?”

Copilot Response:

1. Test the connection to the data source directly (e.g., SQL Server).
2. Check if the data source has any query timeouts or performance issues.
3. Look at recent data source logs for errors or delays.

Step 3:

Address the Gateway : "The data source is fine. Could the issue be with the gateway? How do I troubleshoot that?”

Copilot Response:

1. Test the connection to the data source directly (e.g., SQL Server).
2. Check if the data source has any query timeouts or performance issues.
3. Look at recent data source logs for errors or delays.

Conditional Prompting

Step 4:

Optimize Long-Running Queries: "The gateway is operational. How do I identify and fix long-running queries in the dataset?”

Copilot Response:

1. Test the connection to the data source directly (e.g., SQL Server).
2. Check if the data source has any query timeouts or performance issues.
3. Look at recent data source logs for errors or delays.

Conditional Prompting

Dynamic Problem Solving: Use when the investigation path depends on the findings of previous steps.

Example: Diagnosing an Azure VM connectivity issue, where identifying a DNS problem leads to DNS-specific troubleshooting.

Complex Scenarios with Multiple Possibilities: Use for tasks that have several potential causes or resolutions, requiring adaptive investigation.

Example: Debugging deployment failures in an ARM template based on specific error messages.

Efficient Resource Use: Use when narrowing down the scope saves time by focusing on the most relevant issues.

Example: Analyzing a Power BI refresh failure by first checking the gateway performance and then moving to DAX queries only if needed.

When to Use Conditional Prompting

Tasks with Fixed Processes: Avoid when the task requires a consistent, predefined sequence without deviations.

Example: Performing a routine compliance checklist or standardized server configuration.

Simple, Non-Branched Tasks: Avoid for straightforward tasks where a single step or prompt resolves the issue.

Example: Resetting a system password or confirming a configuration setting.

Limited Context or Incomplete Data: Avoid when critical information is missing, as branching decisions rely on accurate and complete inputs.

Example: Troubleshooting with vague error descriptions or partial logs.

When Not to Use Conditional Prompting

Conditional prompting is ideal for dynamic, context-sensitive workflows that benefit from branching and flexibility. However, it is less suitable for rigid, simple, or highly standardized tasks where fixed processes are more effective.

"Which scenario is ideal for applying Conditional Prompting?"

A. Debugging a Power BI refresh issue by following a fixed series of steps to optimize data source, DAX queries, and visuals.

B. Diagnosing an Azure VM connectivity issue where the next step depends on whether the problem is with NSG rules, VNet configuration, or DNS settings.

C. Investigating a system failure where all potential causes are explored simultaneously.

D. Providing step-by-step instructions for a clearly defined task with no branching paths.

60 Seconds

ITERATIVE PROMPTING

Iterative prompting is a technique where you refine and improve prompts over multiple iterations to guide the LLM toward a better and more accurate response.

Iterative Prompting

Test Customer Query

An Azure Virtual Machine (VM) fails to connect to the network.

The technical support engineer uses iterative prompts to identify and resolve the issue step-by-step.

Iterative Prompting

Step 1: Initial Prompt

Prompt: " "My Azure VM cannot connect to the network. What could be the possible reasons?”

Copilot Response:

1. Incorrect Network Security Group (NSG) rules.
2. Misconfigured Virtual Network (VNet).
3. DNS issues.
4. Missing public or private IP configuration."

Iterative Prompting

Step 2 : Refining for Specific Guidance

Prompt : " "How do I check if Network Security Group (NSG) rules are causing the issue?”

Copilot Response:

1. Use the Azure portal or CLI to view inbound and outbound rules.
2. Ensure the required ports (e.g., RDP: 3389, HTTP: 80, HTTPS: 443) are open.
3. Confirm there are no blocking rules overriding allow rules."

Step 3 : Narrowing Focus

Prompt : "I fixed the NSG rules to allow port 3389. The VM still cannot connect. What should I check next?”

Copilot Response:

1. Ensure the VM is correctly associated with the intended VNet and subnet.
2. Check the subnet route table for misconfigured routes.
3. If using VNet peering, confirm it is active and configured correctly."

Step 4 : Confirming Fix

Prompt : ”The VM is now in the correct subnet, but the issue persists. Could DNS settings or IP configuration be causing the problem?”

Copilot Response:

1. Ensure the VM has a valid private IP assigned.
2. Verify the DNS server configuration in the VNet.
3. Test connectivity using the VM’s private IP and ensure proper name resolution.

Refining Vague or Incomplete Responses: Use when the initial response lacks sufficient detail or misses key elements.

Example: Refining prompts to optimize DAX queries in Power BI after receiving a generic suggestion.

Complex Problem Exploration: Use for tasks requiring detailed exploration and iterative adjustments to reach an optimal solution.

Example: Troubleshooting an Azure deployment issue by iteratively checking configurations, dependencies, and logs.

Creative or Open-Ended Tasks: Use when generating or improving creative outputs such as brainstorming ideas or drafting content.

Example: Iteratively refining the structure of a blog post or designing user interface elements.

When to Use Iterative Prompting

Tasks with Clear or Fixed Outputs: Avoid when a single prompt can sufficiently handle the task without requiring refinements.

Example: Generating a simple SQL query or retrieving static information.

Time-Sensitive or Urgent Tasks: Avoid when multiple iterations could delay the resolution of time-critical problems.

Example: Quickly identifying an obvious error in server configuration.

Highly Structured Workflows: Avoid when tasks follow predefined steps where iteration adds unnecessary complexity.

Example: Running a predefined script for system diagnostics.

When Not to Use Iterative Prompting

Iterative prompting is best suited for refining vague outputs, exploring complex problems, and improving creative tasks. It is less effective for simple, time-sensitive, or highly structured workflows where straightforward approaches suffice.

"In which scenario is iterative prompting most useful?"

A. When investigating a Power BI performance issue by following a predefined sequence of steps.

B. When refining a vague initial response from ChatGPT to achieve a detailed and actionable answer.

C.When troubleshooting a dynamic problem with branching possibilities, such as Azure VM connectivity.

D. When listing all possible solutions for a known technical error.

60 Seconds

Q & A

BREAK (7 MINS)

A reusable, structured framework for crafting effective prompts when interacting with large language models (LLMs).

Prompt Template

Scenario: Diagnosing a Power BI refresh failure. "Explain step-by-step why [issue description] might occur and provide potential solutions for [specific context]."

Filled Template: "Explain step-by-step why a Power BI dataset might fail to refresh and provide potential solutions for gateway configuration and data source connectivity."

Simple Template Example

Advantages of Creating Prompt Templates

Saves time by reusing well-structured prompts.

Ensures consistency in troubleshooting and response quality.

Makes interactions with LLMs more systematic and effective.

Steps to Create Effective Prompt Templates

01

Identify Frequent Issues:

- List recurring customer problems or support scenarios.

02

Determine Key Information:

- Define what the LLM needs to address the issue effectively.

03

Structure the Template:

- Write a clear, reusable prompt format with placeholders for dynamic inputs.

04

Test the Template:

- Run the template on real-world examples and refine for clarity and accuracy.

How To Structure Template

Define Primary Prompts:	Identify core questions or instructions that drive the chain. Ensure prompts are clear, simple, and actionable.
Link Inputs and Outputs:	Determine the information each prompt requires and the outputs it produces. Ensure inputs and outputs logically connect to maintain context.
Implement the Prompt Chain:	Combine templates and primary prompts to create the full sequence. Verify logical progression between prompts.
Test Thoroughly:	Run the prompt chain with sample users or scenarios. Identify areas where prompts may need improvement or clarification.
Iterate and Refine:	Use feedback to adjust prompts, add missing steps, or streamline the sequence.

"What is the primary purpose of a prompt library?"

A. To store predefined prompts for reuse and consistency.

B. To randomly generate prompts for troubleshooting.

C. To experiment with creative, unstructured prompts.

D. To eliminate the need for structured problem-solving.

60 Seconds

"What is the most important factor when building a prompt library?"

A. Including as many prompts as possible, regardless of quality.

B. Ensuring prompts are clear, adaptable, and relevant to recurring issues.

C. Making prompts highly generic to cover all possible use cases.

D. Using prompts only for creative tasks, not structured workflows.

60 Seconds

Take Home Exercise

1. **Activity**: Choose a technical support scenario you frequently encounter (e.g., diagnosing an Azure VM issue, Power BI performance issue).
2. Create a prompt template based on CoT principles.
3. Test your template with a partner or in a small group.
4. Discussion: Share your templates and feedback on how they could be improved.

Q & A